

Project

Consider the three-element system to solve the differential equation with the source term $f(x)$,

$$\frac{d^2u}{dx^2} - 2u = f(x) \quad 0 < x < 1$$
$$f(x) = 4x^2 - 2x - 4$$

subject to the following boundary conditions:

(i) Dirichlet boundary conditions:

$$u = 0 \quad \text{at } x = 0$$
$$u = -1 \quad \text{at } x = 1$$

(ii) Neumann boundary conditions:

$$u = 0 \quad \text{at } x = 0$$
$$\frac{du}{dx} = -3 \quad \text{at } x = 1$$

whose exact solution is given by $u = -2x^2 + x$.

- (a) Solve the above-mentioned problem with FDM, FEM, FVM for both (i) & (ii)
- (b) Prepare a report with all mathematical formulations and results.
- (c) Program the same and submit along with your report.